Practical 4

Implementation and Time Analysis of Factorial Program Using Iterative and Recursive Methods

1. Introduction

The factorial of a non-negative integer n is the product of all positive integers from 1 to n. It is represented by n!.

Formula:

- 0!=1
- $n!=n\times(n-1)\times(n-2)\times\cdots\times1$

Factorials are widely used in mathematics, permutations, combinations, probability, and algorithm design.

2. Aim

To implement the factorial of a number using both iterative and recursive methods and analyze their time and space complexities.

3. Objective

- To understand iterative and recursive approaches.
- To compute the factorial of a given number.
- To compare the performance of both methods. □ To analyze their time and space complexities.

4. Iterative Method

Algorithm

1. Read the number n.
2. Initialize fact = 1.
3. Repeat from 1 to n.
4. Multiply fact by the current number.

5. Print the factorial.

Pseudocode

```
START
Read n
fact = 1

FOR i = 1 TO n
   fact = fact × i
ENDFOR

Print fact
STOP
```

5. Recursive Method

Algorithm

1. Read the number n.
2. If n is 0 or 1, return 1.
3. Otherwise return n × factorial(n−1).
4. Print the factorial.

Pseudocode

```
START

FUNCTION factorial(n)

IF n == 0 OR n == 1
   RETURN 1
ELSE
   RETURN n × factorial(n-1)
ENDIF

Read n
```

Print factorial(n)

STOP

Code:

Using Iterative Method

```
n = int(input("Enter a number: "))

fact = 1

for i in range(1, n + 1):
    fact *= i

print("Factorial =", fact)
```

```
Enter a number: 5
Factorial = 120
```

Using Recursive Method

```
def factorial(n):
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)

n = int(input("Enter a number: "))

print("Factorial =", factorial(n))
```

```
Enter a number: 6
Factorial = 720
```

6. Time and Space Complexity Comparison

Method	Best Case	Average Case	Worst Case	Space Complexity
Iterative Factorial	O(n)	O(n)	O(n)	O(1)
Recursive Factorial	O(n)	O(n)	O(n)	O(n)

7. Advantages

Iterative Method

- Easy to implement.
- Uses constant memory.
- Faster than recursion due to no function call overhead.

Recursive Method

- Simple and elegant code.
- Easy to understand recursive concepts.
- Suitable for recursive mathematical problems.

8. Disadvantages

Iterative Method

- Code can become lengthy for complex recursive problems. □ Less intuitive for recursive algorithms.

Recursive Method

- Uses extra memory due to the function call stack.
- May cause stack overflow for very large values of n.

9. Conclusion

The factorial of a number can be calculated using both iterative and recursive methods. Both approaches require O(n) time. However, the iterative method is more memoryefficient with O(1) space complexity, while the recursive method requires O(n) space because of recursive function calls. Therefore, the iterative approach is generally preferred for computing factorials, especially for large input values.